Essential Summary

Generalized Padé Approximation Method for Homoclinic Orbits of Strongly Nonlinear Autonomous Oscillators

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1 Background and Motivation

The determination of homoclinic orbits is central to understanding global bifurcations, chaos, and solitary wave solutions in nonlinear dynamics [1]. Existing Padé-based approaches share a common limitation: the numerator and denominator of the approximant are restricted to polynomial functions, which limits flexibility in capturing the exponential decay behavior characteristic of homoclinic orbits.

This paper introduces the **Generalized Padé Approximation (GPA) method**, a theoretical extension that removes this restriction by allowing the approximant to be constructed from *any* kind of function series. This provides a unified framework that unifies all existing Padé-type modifications as special cases.

2 Core Innovation: The Generalized Padé Approximation

2.1 From Classical to Generalized

The classical Padé approximant of a formal power series $f(z) = \sum_{i=0}^{\infty} a_i z^i$ is a rational function satisfying:

$$f(z) - \frac{P_L(z)}{Q_M(z)} = \mathcal{O}(z^{L+M+1}), \qquad Q_M(0) = 1. \tag{1}$$

The generalized definition extends H_m to $\widehat{H}_m$:

$$\boxed{\widehat{H}_m = \left\{ P : P(z) = \sum_{i=0}^{m} a_i g_i(z)^i,\ g_i : \mathbb{C} \to \mathbb{C},\ a_i \in \mathbb{C} \right\}}, \tag{2}$$

and $G(m,n) = \{R = P/Q : P \in \widehat{H}_m,\ Q \in \widehat{H}_n \setminus \{0\}\}$. The **GPA**$[L/M]_f$ is then the rational function $P_L/Q_M \in G(L,M)$ satisfying the same $\mathcal{O}(z^{L+M+1})$ condition. When $g_i(z) = z^i$, the GPA reduces to the classical Padé approximant.

2.2 Homoclinic Orbit Construction via Hyperbolic Functions

For $\ddot{x} + g(x) = 0$, the power series solution $x(t) = \sum_{i=0}^{\infty} a_i t^i$ is first obtained. For a homoclinic orbit through saddle $H(h, 0)$, the initial amplitude a and initial velocity ($a_1 = 0$) are determined by the energy integral $\int_h^a g(x)\,dx = 0$.

Since $\lim_{t\to\pm\infty} x(t) = h$, a power series cannot capture the asymptotic behavior. The key innovation — construct a GPA using **hyperbolic functions** that naturally satisfy the boundary conditions:

$$\boxed{\mathrm{GPA}[L/M] = \frac{\sum_{i=0}^{L} \alpha_i \operatorname{sech}^i t}{1 + \sum_{i=1}^{M} \beta_i \operatorname{sech}^i t}}. \tag{3}$$

The $L+M+1$ unknown coefficients $\{\alpha_i, \beta_i\}$ are determined by expanding the GPA into a Taylor series around $t=0$ and matching coefficients with the power series $\{a_i\}$. The boundary condition $x(\pm\infty) = h$ is automatically satisfied ($\operatorname{sech}^i t \to 0$). The coefficient matching reduces to a system of $L+M$ linear equations coupled with one nonlinear condition from the energy integral.

2.3 Advantages

The $\operatorname{sech}^i t$ basis offers three advantages: (1) its Maclaurin series is simpler than exponentials, reducing algebraic effort when matching terms; (2) it satisfies $x(\pm\infty) = h$ automatically; (3) it applies to *any* $g(x)$ that is a polynomial or rational function.

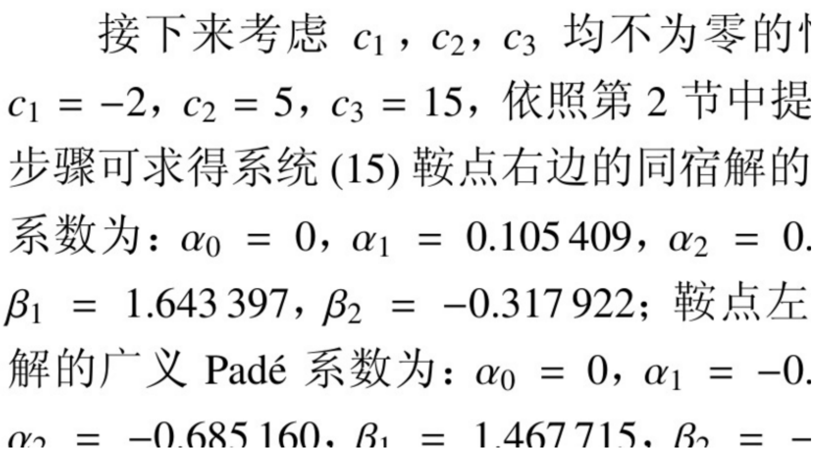
3 Key Results: Three Numerical Examples

3.1 Example 1: Strongly Asymmetric Helmholtz–Duffing

$$\ddot{x} + c_1 x + c_2 x^2 = 0, \qquad c_1 = -3,\ c_2 = 100. \tag{4}$$

Despite the enormous asymmetric nonlinearity ($c_2 = 100 \gg 1$), GPA[2/2] matches Runge–Kutta with excellent agreement:

$$\mathrm{GPA}[2/2] = \frac{0.005 \operatorname{sech} t + 0.07 \operatorname{sech}^2 t}{1 + 0.611111 \operatorname{sech} t + 0.055555 \operatorname{sech}^2 t}. \tag{5}$$



接下来考虑 c_1，c_2，c_3 均不为零的
$c_1 = -2$，$c_2 = 5$，$c_3 = 15$，依照第 2 节中提
步骤可求得系统 (15) 鞍点右边的同宿解的
系数为：$\alpha_0 = 0$，$\alpha_1 = 0.105\,409$，$\alpha_2 = 0.$
$\beta_1 = 1.643\,397$，$\beta_2 = -0.317\,922$；鞍点左
解的广义 Padé 系数为：$\alpha_0 = 0$，$\alpha_1 = -0.$
$\alpha_2 = -0.685\,160$，$\beta_1 = 1.467\,715$，$\beta_2 = -$

Figure 1: **Figure 1.** (Original Fig. 1) Homoclinic orbit of the Helmholtz–Duffing oscillator with $c_1 = -3, c_2 = 100$. Solid: Runge–Kutta. Dotted: GPA[2/2]. The method maintains accuracy despite the strongly asymmetric nonlinearity.

3.2 Example 2: Full Cubic–Quadratic Helmholtz–Duffing

$$\ddot{x} + c_1 x + c_2 x^2 + c_3 x^3 = 0, \qquad c_1 = -2,\ c_2 = 5,\ c_3 = 15. \tag{6}$$

Both left and right homoclinic orbits are obtained simultaneously with $L = M = 2$, demonstrating the method handles asymmetric cubic–quadratic nonlinearities without modification.

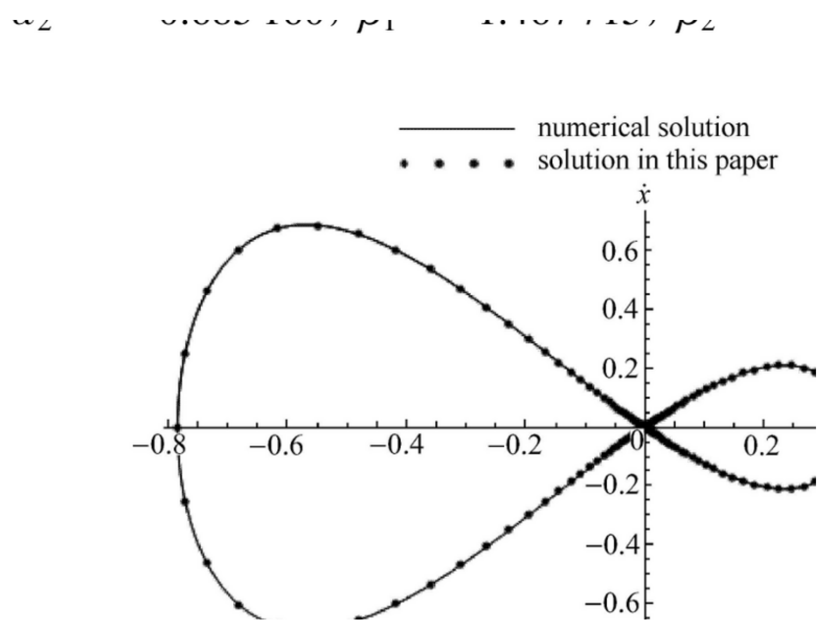


Figure 2: **Figure 2.** (Original Fig. 2) Homoclinic orbits for $c_1 = -2, c_2 = 5, c_3 = 15$. Solid: Runge–Kutta. Dotted: GPA[2/2], capturing both left and right homoclinic orbits.

3.3 Example 3: Duffing–Harmonic Oscillator (Rational Restoring Force)

$$\ddot{x} + \frac{\lambda x + \mu x^3}{1 + \nu x^2} = 0, \qquad \lambda = -1,\ \mu = 1,\ \nu = 1. \tag{7}$$

With $L = M = 3$, the GPA successfully captures homoclinic orbits of this rational-restoring-force oscillator. The exact solution cannot be expressed in elementary or classical elliptic functions — yet the GPA handles it without modification.

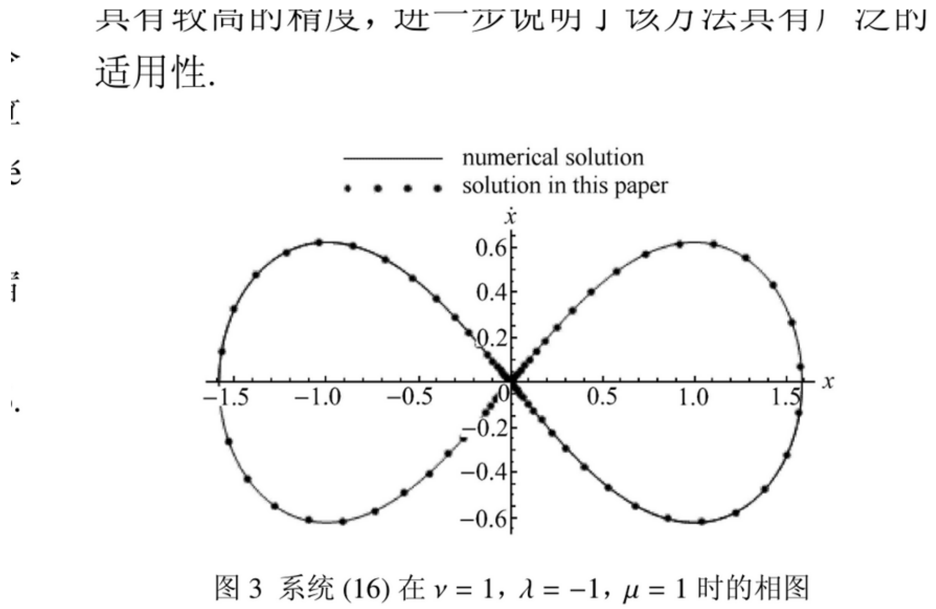


Figure 3: **Figure 3.** (Original Fig. 3) Homoclinic orbits of the Duffing-harmonic oscillator with $\lambda = -1, \mu = 1, \nu = 1$. Solid: Runge–Kutta. Dotted: GPA[3/3]. The method handles rational restoring forces without additional computational complexity.

4 Significance and Impact

1. **Theoretical unification:** The GPA definition unifies all existing Padé-type modifications (quasi-Padé, exponential-function approximants, modified Padé) under a single theoretical framework.

2. **Computational efficiency:** The $\operatorname{sech}^i t$ basis produces simpler Taylor expansions than exponential-based alternatives, reducing the cost of coefficient matching at the same approximation order.

3. **Broad class coverage:** The method handles polynomial $g(x)$ of arbitrary degree and rational $g(x)$ — a significant advantage over methods restricted to specific restoring force forms.

5 Limitations

- Conservative systems only ($\varepsilon = 0$); damped oscillators require separate methods.
- Approximation order $L = M$ chosen empirically; no a priori error estimate.
- Single-degree-of-freedom systems only.

6 Connection to the GHFP/MGHFP Series

This paper (together with its English version, Chin. Phys. B 2014) and the GHFP/MGHFP series represent two **complementary methodological families**. Both aim at analytical treatment of strongly nonlinear oscillators, but via different mathematical routes: GPA constructs homoclinic/heteroclinic solutions directly through rational functions of hyperbolic functions in the time domain; GHFP/MGHFP uses perturbation theory in the angular domain to derive amplitude–parameter relationships. The $\operatorname{sech}^i t$ and $\tanh^i t$ basis functions introduced here also appeared in the GHFP framework (JSV 2013), highlighting the shared mathematical intuition across both families.

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References

[1] A.H. Nayfeh and B. Balachandran, *Applied Nonlinear Dynamics*, Wiley, 1995.

[2] Z. Li, J. Tang, and P. Cai, "A generalized Padé approximation method of solving homoclinic and heteroclinic orbits of strongly nonlinear autonomous oscillators," *Chin. Phys. B*, vol. 23, no. 12, p. 120501, 2014.